

Hands-On-Learning: Use VirusTotal,
Any.Run, and Shodan for Malware and
Infrastructure Research

Learning Objectives

By the end of this activity, learners will be able to collect data artifacts, IOCs and TTPs, analyze the data, and map a historical campaign with attribution in order to create a report for management.

Description

Use VirusTotal, Any.Run, and Shodan for Malware and Infrastructure Research

Structure of the Activity

Title: Use VirusTotal, Any.Run, and Shodan for Malware and Infrastructure Research

Tools used: VirusTotal, Any.Run, and Shodan

Scenario

Using the tools and skills you've developed in this course so far, investigate and map a historical cyber-attack campaign, including IOCs and TTPs.

Objective

Instructions for learners:

Use VirusTotal, Any.Run, Shodan for malware and infrastructure research

Investigate and map a historical campaign

Use VirusTotal, ANY.RUN, and Shodan to analyze artifacts, identify indicators, and map infrastructure related to a historical threat campaign (e.g., Emotet, APT29, SolarWinds, TrickBot).

Tools:

VirusTotal

ANY.RUN

Shodan

STEP 1: Select a Campaign to Investigate

Choose a known, documented campaign:

Use: MITRE ATT&CK, CISA Alerts, threat reports (e.g., “TrickBot Infrastructure 2022”)

Example: Emotet Return Campaign (late 2021–2022)

Goal: Get starting IOCs (hashes, domains, IPs)

STEP 2: Investigate Samples with VirusTotal

Find a known malware hash (from report or ANY.RUN sandbox)

Go to VirusTotal

Paste the file hash into the search bar

What to Analyze:

Behavior tab: Network connections, dropped files

Relations tab: Related URLs, domains, files

Communicating files/IPs: Note indicators (C2, payload sites)

Graph: Build a pivotable view of interconnected assets

Capture:

2–3 domains or IPs

1–2 dropped files or URLs

Screenshot of the graph view

STEP 3: Explore Behavior with ANY.RUN

Visit ANY.RUN (Free account or public sandbox)

Search for a file hash or malware family (e.g., Emotet)

View interactive behavior:

Network traffic

Processes tree

Suspicious actions (e.g., PowerShell, reg keys)

What to Capture:

Command and control (C2) domains/IPs

Payload download URLs

Execution flow & persistence mechanisms

Take notes on:

TTPs (Map to MITRE if possible)

DNS traffic, HTTP POST patterns

STEP 4: Scan Infrastructure with Shodan

Go to Shodan.io

Search for a C2 IP address or domain from earlier steps

Investigate:

Open ports

Banner data (web server type, RDP, etc.)

Location, ISP, last seen

Optional:

Use advanced filters:

port:443 org:"Contabo" (if Contabo is hosting C2)

ssl.cert.subject.CN:"example.com"

Take notes on:

Hosting providers

Clusters of related IPs

Patterns in infrastructure reuse

STEP 5: Map the Campaign

Use your findings to build a simple diagram or report:

Initial payload hash

Dropper → Payload → C2

Infrastructure (IP, ports, geography)

TTPs used (execution, persistence, C2, exfiltration)

Optional Tools:

Draw.io

MISP or MITRE ATT&CK Navigator

PowerPoint or Lucidchart

Lab Deliverables:

Task Evidence

Campaign selected Name & brief summary

VirusTotal scan IOC list + graph screenshot

ANY.RUN sandbox Behavior + C2 notes

Shodan scan IP insights (port, banner, ASN)

Infrastructure map Diagram or structured table

MITRE mapping At least 2–3 TTPs

Organize Information Following This Structure for Peer Review

- 1 Title Page:** Project title, your name, date.
- 2 Requirement 1:** Provide details here.
- 3 Requirement 2:** Provide details here.
- 4 Requirement 3:** Provide details here.

Instructions for Documenting and Sharing the Project for Peer Review



1. Document the Project

Use word processing software to create your report.

Insert the diagram, map, screenshot, or image directly into your Word document, if available.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



2. Share the Project

Save your report and presentation in PDF format.

Upload your documents to the “Peer Review” area in the course shell.

Provide a brief description of your project in the submission post.



3. Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.

This activity gives learners a chance to apply the lesson by ...